FIN 522A: Lecture 2

January 13, 2026

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Yield Curve Construction, Pricing, and Interest-Rate Risk

- In this lecture we will discuss how to:
  - Construct a pure curve from market instruments
  - Use the curve to price bonds
  - Measure interest-rate risk (duration/convexity)

Why This Matters

- Markets price fixed income off a curve
- Risk is movement of the curve
- Hedging is matching sensitivities to the curve

What You'll Be Able to Do

- Bootstrap discount factors
- Convert to spot and forward rates
- Price a bond using the curve
- Compute Macaulay duration, modified duration, dollar duration

Core Pricing Identity

- To refresh:

$$P = \sum_{t=1}^{T} CF_t \cdot DF(t)$$

- $\mathrm{CF_t}$ are contractual (known)
- $\mathrm{DF(t)}$ are market-implied (constructed)
- Everything else is derived

What Moves Prices

- Cash flows generally don’t change (plain-vanilla bonds)
- Discount factors change when:
  - the risk-free curve moves
  - spreads move
  - liquidity premia move

Yield-to-Maturity Is Not a Pricing Curve

- YTM solves one discount rate that equates PV to price
- But bonds have multiple maturities of cash flow
- We need $\mathrm{DF}(1)$, $\mathrm{DF}(2)$, … not one YTM
- Just to refresh, here is YTM formula:

$$P = \sum_{t=1}^{nm} \frac{\frac{C}{m}}{\left(1+\frac{y}{m}\right)^t} + \frac{F}{\left(1+\frac{y}{m}\right)^{nm}}$$

P = current market price of the bond

C = **annual** coupon payment (in currency units, not a rate)

F = face (par) value of the bond

y = annualized yield to maturity

m = number of coupon payments per year

n = years to maturity

t = time index for coupon payment periods, $t = 1, 2, \ldots, nm$

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Discount Factors, Spot Rates & Bootstrapping

- $\mathrm{DF(t)}$ is the price today of \$1 received at time t
- Pure pricing objects; yields are summaries

$$DF(t) = \frac{1}{(1+z_t)^t}$$

- z_t: spot rate to maturity t
- Given $\mathrm{DF(t)}$, you can compute z_t
- Given z_t, you can compute $\mathrm{DF(t)}$

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Why You Can't "Just Use the Yield Curve"

- Market quotes are usually:
  - yields on specific instruments (coupon bonds)
  - swap rates
- A pricing curve must be:
  - internally consistent (no-arbitrage)
  - expressed as $\mathrm{DF}(t)$, z_t, and implied forwards

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Market-Observed Yield Curves

- Instrument-Specific Yield Curves
  - Curves constructed from prices of specific traded instruments
  - Examples:
    - On-the-run Treasury yields
    - Swap rate curves
    - Agency bond curves
  - Typically quoted as:
    - yields
    - par rates

On-the-Run vs Off-the-Run Securities

On-the-Run Instruments

- Most recently issued selling at or near par value
- Highest liquidity
- Tightest bid–ask spreads
- Often trade at a liquidity premium (lower yields)

Off-the-Run Instruments

- Older issues
- Lower liquidity
- Wider bid–ask spreads
- May trade at higher yields despite identical cash flows

Yield Curves published in the financial press are typically on-the-run curves

From Instrument Prices to a Pricing Curve

- Markets give you: prices of bonds, swaps, bills, futures
- Pricing requires:
  - discount factors $DF(t)$
  - spot rates zt
  - implied forward rates
- The curve is inferred — not observed.

Dreams vs Reality

Ideal Inputs: Zero-Coupon Bonds

- One cash flow → one discount factor directly
- Not always available at all maturities

Real Inputs: Coupon Bonds

- Multiple cash flows → multiple maturities bundled
- Must infer discount factors indirectly

Treasury STRIPS (Bridging Ideal and Real Inputs)

- STRIPS = Separate Trading of Registered Interest and Principal of Securities
  - U.S. Treasury program that allows payments and principal to trade as separate zero-coupon securities
  - Why STRIPS Matter for Yield Curves:
    - Each STRIP is a true zero-coupon instrument
    - STRIPS provide direct observations of discount factors (do not bundle multiple cashflows)
  - Limitation of STRIPS:
    - Not available at all maturities
    - Liquidity varies across maturities
    - Still subject to supply/demand imbalances as well as repo and financing effects

STRIPS as a tool

- STRIPS are the closest real-world approximation to ideal zero-coupon inputs
- Even STRIPS require:
  - curve construction
  - Interpolation
  - no-arbitrage enforcement
- STRIPS get us close to the ideal. The pure curve formalizes this idea across all instruments

Bootstrapping to a Pure Curve

- Use shortest maturity instruments first
- Each instrument adds:
  - one equation
  - one new unknown $\mathrm{DF(t)}$
- $\mathrm{DF(3)}$ depends on $\mathrm{DF(1)}$ and $\mathrm{DF(2)}$
- Early error will contaminate later curve

Bootstrapping Example

- Assumptions: annual coupons; face value = 100; t = 1,2,3
- Observed market instruments:
  - 1Y zero price = 97
  - 2Y coupon bond: coupon 4%, price = 100
  - 3Y coupon bond: coupon 5%, price = 102
- Goal: compute $\mathrm{DF}(1), \mathrm{DF}(2), \mathrm{DF}(3)$

Bootstrapping Example (con.)

- Step 1:
  - Solve for $\mathrm{DF}(1)$:
    - $97 = 100 \cdot \mathrm{DF}(1)$
    - $\mathrm{DF}(1) = 0.97$
  - Then convert to 1Y spot rate:
    - $\mathrm{DF}(1) = 1 / (1 + z_1)$
    - $1 + z_1 = 1 / 0.97$
    - $z_1 \approx 3.09\%$

Bootstrapping Example (con.)

- Step 2:
  - Write 2Y Bond Cash Flows:
    - 2Y 4% coupon bond cash flows:
      - t=1: 4
      - t=2: 104
    - Pricing equation:
      - $100 = 4 \cdot \mathrm{DF}(1) + 104 \cdot \mathrm{DF}(2)$
  - Plug in $\mathrm{DF}(1)$:
    - $100 = 4(0.97) + 104 \cdot \mathrm{DF}(2)$
    - $4(0.97) = 3.88$
    - $100 = 3.88 + 104 \cdot \mathrm{DF}(2)$

Bootstrapping Example (con.)

- Solve for $\mathrm{DF}(2)$:
  - $104 \cdot \mathrm{DF}(2) = 100 - 3.88 = 96.12$
  - $\mathrm{DF}(2) = 96.12 / 104 = 0.9242308$
  - $\mathrm{DF}(2) \approx 0.92423$
- Convert $\mathrm{DF}(2)$ to a 2Y spot rate:
  - $\mathrm{DF}(2) = 1 / (1 + z_2)^2$
  - $(1 + z_2) = \mathrm{DF}(2)^{(-1/2)}$
  - $z_2 \approx 4.02\%$

Bootstrapping Example (con.)

- And so forth- Step 3:
  - Write 3Y 5% coupon bond cash flows:
    - t=1: 5
    - t=2: 5
    - t=3: 105
  - Pricing equation:
    - 102 = $5\mathrm{DF}(1)$ + $5\mathrm{DF}(2)$ + $105\mathrm{DF}(3)$
  - Plug in $\mathrm{DF}(1)$ and $\mathrm{DF}(2)$:
    - 102 = 5(0.97) + 5(0.9242308) + $105\mathrm{DF}(3)$
    - 5(0.97) = 4.85
    - 5(0.9242308) = 4.621154
    - 102 = 9.471154 + $105\mathrm{DF}(3)$

Bootstrapping Example (con.)

- Solve $\mathrm{DF}(3)$:
  - $105\mathrm{DF}(3) = 102 - 9.471154 = 92.528846$
  - $\mathrm{DF}(3) = 92.528846 / 105 = 0.8812271$
  - $\mathrm{DF}(3) \approx 0.88123$
- Convert $\mathrm{DF}(3)$ to a 3Y Spot Rate
  - $\mathrm{DF}(3) = 1 / (1 + z_3)^3$
  - $(1 + z_3) = \mathrm{DF}(3)^{(-1/3)}$
  - $z_3 \approx 4.31\%$

Summary: The Pure Curve We Built

- Discount factors:
  - $\mathrm{DF}(1)$ = 0.97
  - $\mathrm{DF}(2)$ ≈ 0.92423
  - $\mathrm{DF}(3)$ ≈ 0.88123
- Spot rates (approximate):
  - z_1 ≈ 3.09%
  - z_2 ≈ 4.02%
  - z_3 ≈ 4.31%

Why On-the-Run Yield Curves ≠ Pure Yield Curves

- Par yields come from coupon instruments
- Treating par yields as zero rates misprices coupon bonds
- Pure curve is constructed to price inputs exactly

Let's look at a common mistake:

- Assume annual coupons, face = 100, par price = 100.
- Quoted par yields:
  - 1Y: 2.00%
  - 2Y: 4.00%
  - 3Y: 5.00%
- Goal: build DF that price these par bonds at 100

Mistake using par yields

- $\mathrm{DF}(n) = 1 / (1 + y_n)^n$
- $\mathrm{DF}(1) = 1/1.02 = 0.980392$
- $\mathrm{DF}(2) = 1/1.04^2 = 0.924556$
- $\mathrm{DF}(3) = 1/1.05^3 = 0.863838$

Mispricing Check

- We can see if we price the 2Y par bond with these DFs:
- P = 4(0.980392) + 104(0.924556)
- P = 3.921568 + 96.154 = 100.0756
- Does not equal 100

Correct: Bootstrap DF(1) from 1Y Par Bond

- 1Y par bond coupon = 2
- 100 = $102\mathrm{DF}(1)$
- $\mathrm{DF}(1)$ = 0.980392

Correct: Bootstrap DF(2)

- 2Y par bond coupon = 4
- 100 = $4\mathrm{DF}(1)$ + 104DF(2)
- Plug $\mathrm{DF}(1)$=0.980392:
- 100 = 3.921568 + $104\mathrm{DF}(2)$
- $\mathrm{DF}(2)$ = 96.078432 / 104 = 0.923831

Correct: Bootstrap DF(3)

- 3Y par bond coupon = 5
- 100 = $5\mathrm{DF}(1)$ + $5\mathrm{DF}(2)$ + $105\mathrm{DF}(3)$
- $5\mathrm{DF}(1)$=4.90196; $5\mathrm{DF}(2)$=4.619155
- Sum = 9.521115
- $105\mathrm{DF}(3)$ = 100 – 9.521115 = 90.478885
- $\mathrm{DF}(3)$ = 0.861704

Spot Rates from Bootstrapped DF

- $z_n = DF(n)^{-1/n} - 1$
- $z_1 = 2.00\%$
- $z_2 \approx 4.04\%$
- $z_3 \approx 5.09\%$

What We Have Now & The Core Risk Question

- A discount curve (state)
- A pricing rule for fixed cash-flow streams
- What happens when the curve moves?
  - Rates move → discount factors move → prices move
- Assume a parallel shift of +Δy to all spot rates:

$$DF'(t) = 1 / (1 + z_t + \Delta y)^t$$

What is more sensitive, short or longer maturities?

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Full repricing vs Approximation

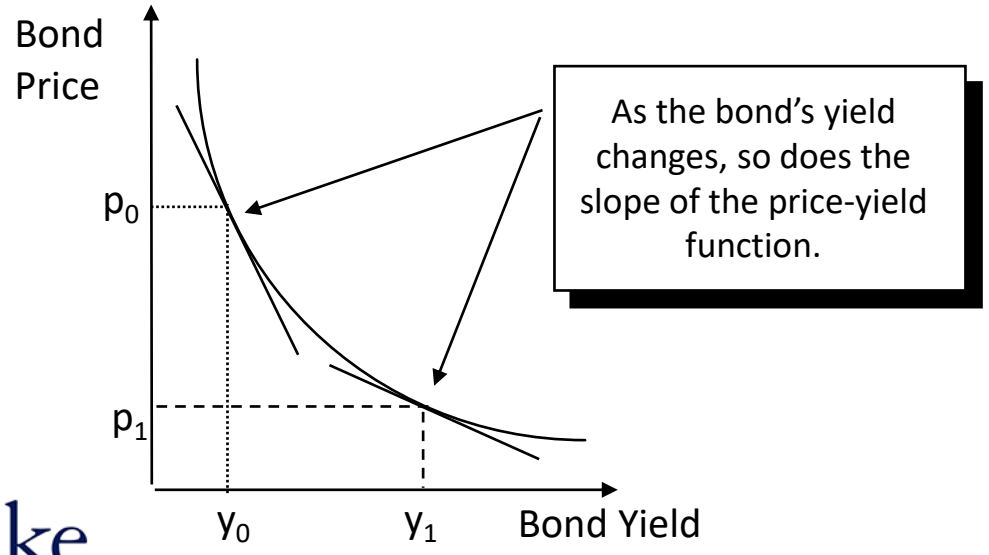
- Full Repricing: this is the truth under the assumed curve movement:

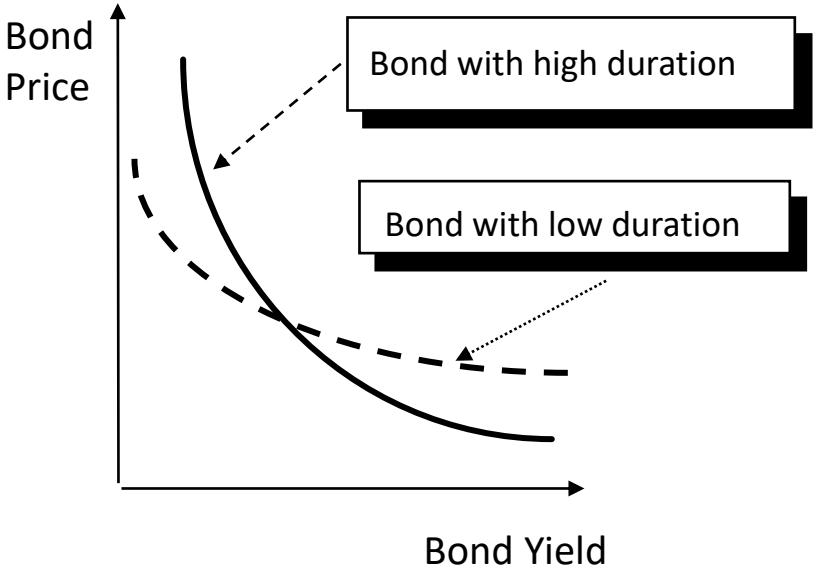
$$P = \sum CF_t \cdot DF(t)$$

- However:
  - Full repricing for every small shock can be costly
  - We may want to simply estimate market value changes
  - Approximations summarize sensitivity: duration, convexity
  - Weighted average portfolio duration provides insight into how an entire portfolio's value will change, based on re-pricing just benchmark bond(s)

Duration

- Duration = effective timing of value
- Higher duration => more price change for a given interest rate change
- Duration at any given yield: estimated as the slope of the price-yield function.
- Duration = $\frac{dP}{dy}$





Why do coupons reduce duration?

- “Effective time to get your money back”
- Zero-coupon bond: all cash flows at duration = maturity
- Coupon bond: duration < maturity (cash flows arrive earlier)
- Coupons shift PV earlier
  - Higher coupon → shorter duration
  - Shorter maturity → shorter duration

Macaulay Duration

$$D_M = \sum_{t=1}^{T} t \cdot \frac{CF_t \cdot DF(t)}{P}$$

- Weights are PV shares:

$$w_t = \frac{CF_t DF(t)}{P}$$

- And $\sum w_t = 1$

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Macaulay Duration Example

- Step 1: Compute PV of Each Cash Flow
  - Assume a 3Y 5% Bond
  - Use curve we calculated in our bootstrapping:
    - $\mathrm{DF}(1)$=0.97; $\mathrm{DF}(2)$=0.92423; $\mathrm{DF}(3)$=0.88123
  - Present Value Components:
    - $PV_1 = 5 \cdot \mathrm{DF}(1) = 5(0.97) = 4.85$
    - $PV_2 = 5 \cdot \mathrm{DF}(2) = 5(0.92423) = 4.62115$
    - $PV_3 = 105 \cdot \mathrm{DF}(3) = 105(0.88123) = 92.5292$
- Step 2: Total Price
  - $P = PV_1 + PV_2 + PV_3$
  - $P \approx 4.85 + 4.62115 + 92.5292 = 102$ (rounded)

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Macaulay Duration Example (con.)

- Step 3: Weights
  - $w_1 = PV_1 / P$
  - $w_2 = PV_2 / P$
  - $w_3 = PV_3 / P$
  - Approx:
  - $w_1 \approx 4.85/101 \approx 0.048$
  - $w_2 \approx 4.62/101 \approx 0.045$
  - $w_3 \approx 92.53/101 \approx 0.907$

Macaulay Duration Example (con.)

- Step 4: Compute
  - $D_M = 1 \cdot w_1 + 2 \cdot w_2 + 3 \cdot w_3$
  - $\approx 1(0.048) + 2(0.045) + 3(0.907)$
  - $\approx 0.048 + 0.092 + 2.718$
  - $D_M \approx 2.859$ years

Modified Duration

- Macaulay measures time; modified measures sensitivity.
- Annual compounding: $D_{mod} = D_M / (1 + y)$ [where y = yield]
- Compounding with a frequency of m times per year: $D_{mod} = D_M / (1 + y/m)$
- Continuous compounding (if y is cont. comp.): $D_{mod} = D_M$

- Compounding matters -> duration depends on quoting convention

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Modified Duration as Price Sensitivity

- Choose a Yield for Modified Duration:
  - If the 3Y bond's yield is ~4.3% (roughly $z3$ as a proxy):
    - $y \approx 4.3\%$
    - Then: $D_{mod} \approx 2.859/1.043 = 2.74$
    - $\Delta P / P \approx - D_{mod} \cdot \Delta y$
    - If $\Delta y = +1\%$:
    - Then $\Delta P / P \approx -2.74\%$

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DV01

- DV01 (per 1 bp):
  - DV01 = $D_{mod} \cdot P \cdot 10^{-4}$
- Using $D_{mod} \approx 2.74$ and $P \approx 102$
  - DV01 = $2.74 \cdot 102 \cdot 10^{-4} \approx 0.027948$
- Interpretation: ~$0.0279 change per $100 face for a 1 bp move

$$\Delta P \approx -(\mathrm{DV01}) \times \Delta y_{\mathrm{bp}}$$

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Applying DV01 Duration Example

The 10yr UST has the following characteristics:

- Price = 102.68
- Yield = 5.81%
- DV01 = 0.07572

Now bond yield goes up by 20bp to 6.01%. What is the new price?

Price change	=	- 20 * 0.07572
	=	- 1.514
New Price	=	102.68 - 1.514
	=	101.17

Dollar Duration vs DV01

- Some sources equate ‘dollar duration’ to mean DV01, but:
  - in market practice DV01 is per basis point (Hull)
  - dollar duration is per percent— So 100 bps
  - the difference is units

Dollar Duration

- Dollar Duration (per 100 bp or 1%):
  - $DD = D_{mod} \cdot P \cdot 10^{-2}$
- Using $D_{mod} \approx 2.74$ and $P \approx 102$
  - $DD = 2.74 \cdot 102 \cdot 10^{-2} \approx 2.79$
- Interpretation: ~$2.79 change per $100 face for a 100 bp move
- How do we think of Δy here? Per 100 bps.

$$\Delta P \approx -(\text{Dollar Duration}) \times \frac{\Delta y_{\text{bp}}}{100}$$

- Note DD can be expressed as $D_{mod} \cdot P$ by convention sometimes – works if Δy is expressed as a decimal, i.e. 100 bps = 0.01

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Applying Dollar Duration to the Example

The 10yr UST has the following characteristics:

- Price = 102.68
- Yield = 5.81%
- Dollar Duration = 7.572

Now bond yield goes up by 20bp to 6.01%. What is the new price?

Price change	=	- (20/100) * 7.572
	=	- 1.514
New Price	=	102.68 - 1.514
	=	101.17

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Further Look At Dollar Duration

- Example:
  - What is the change in market value of $1MM par amount (or face value) of the 5yr US Treasury for a 15bp increase in yield?
  - The 5yr UST has the following characteristics:
    - Price = 102.68
    - DD = 3.368
- Hence, the change in market value of $1MM for a 15bp change in yield :

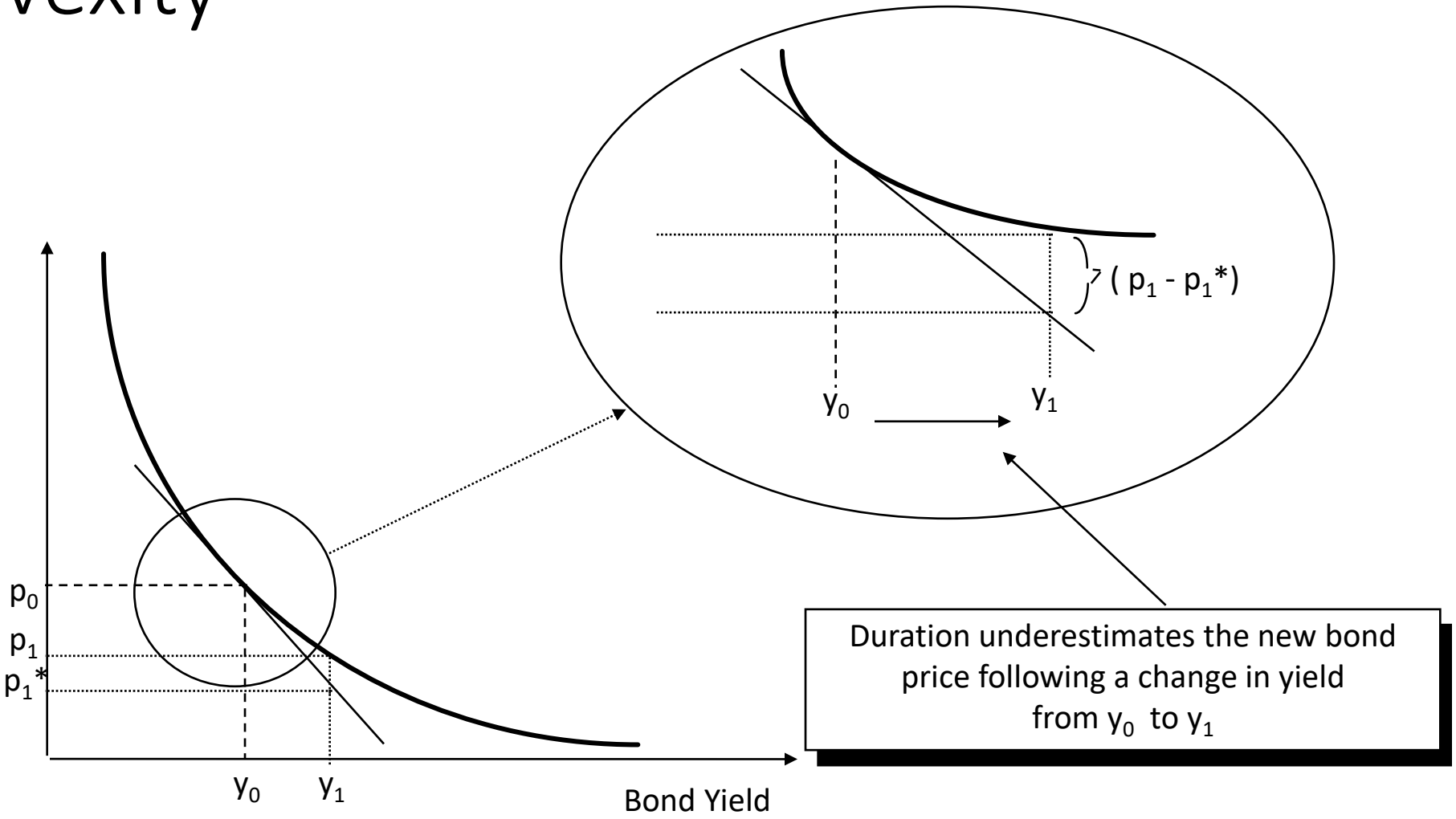
$$= 3.368 *(15/100) * 10{,}000 = \$5{,}052$$

- Why are we multiplying by 10,000 here (after all, the 'par' or 'face' value that we own is 1,000,000, not 10,000)?

Why Duration Is Not Enough

- Duration is linear (first-order)
- Price-yield relationship is curved
- Next: convexity improves the approximation
  - Convexity captures curvature of the price–yield relationship
  - Improves accuracy for larger rate moves
  - Particularly important for long maturities

Convexity



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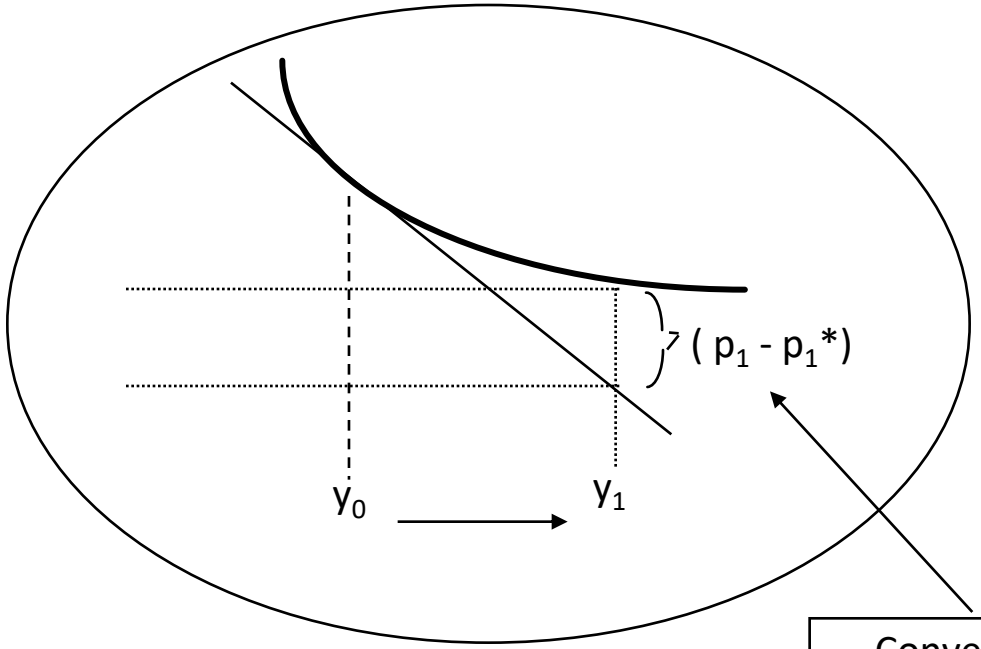
Improving On The Duration Estimate: Convexity

Convexity: a first approximation to the duration measurement error. A bond's convexity is a measure of the "curved-ness" of the price/yield function.



Convexity estimates the distance between p_1 and p_1*

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Convexity (Discrete Compounding): Derivation + Worked Example

- Why Convexity Exists
  - Duration is a first-order (linear) approximation of price–yield sensitivity
  - Convexity adds the second-order (curvature) term so the approximation remains usable for larger moves
- Step 1 — Start from the bond price as a function of yield (annual compounding)

$$P(y) = \sum_{t=1}^{T} \frac{CF_t}{(1+y)^t}$$

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Convexity Derivation (con.)

- Step 2 — First derivative (duration comes from this)

$$\frac{dP}{dy} = \sum_{t=1}^{T} -t \cdot CF_t \cdot (1+y)^{-(t+1)}$$

- Divide by P to get the familiar approximation:

$$\frac{\Delta P}{P} \approx -D_{mod} \Delta y$$

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Convexity Derivation (con.)

- Step 3 — Second derivative (this is convexity)

$$\frac{d^2P}{dy^2} = \sum_{t=1}^{T} t(t+1) \cdot CF_t \cdot (1+y)^{-(t+2)}$$

- Define (discrete) convexity:

$$C \equiv \frac{1}{P}\frac{d^2P}{dy^2} = \frac{1}{P}\sum_{t=1}^{T} \frac{t(t+1)\,CF_t}{(1+y)^{t+2}}$$

- Then the second-order price approximation is:

$$\frac{\Delta P}{P} \approx -D_{mod}\Delta y + \frac{1}{2}C(\Delta y)^2$$

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Convexity of a 3Y 5% Annual Coupon Bond at y = 5%

- Assume face = 100, annual coupons, so cash flows:
  - CF_1=5, CF_2=5, CF_3=105
- At y=5%, this bond is at par: $P = \frac{5}{1.05} + \frac{5}{1.05^2} + \frac{105}{1.05^3} = 100$
- Compute convexity using: $C = \frac{1}{P}\sum_{t=1}^{3}\frac{t(t+1)CF_t}{(1+y)^{t+2}}$
- Term-by-term (annual compounding):
  - $t = 1: \frac{1 \cdot 2 \cdot 5}{1.05^3} = \frac{10}{1.157625} = 8.6384$
  - $t = 2: \frac{2 \cdot 3 \cdot 5}{1.05^4} = \frac{30}{1.215506} = 24.6811$
  - $t = 3: \frac{3 \cdot 4 \cdot 105}{1.05^5} = \frac{1260}{1.276282} = 987.2430$

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Convexity of a 3Y 5% Annual Coupon Bond at y = 5% (con.)

- Sum:

$$\sum = 8.6384 + 24.6811 + 987.2430 = 1020.5624$$

- Divide by P=100: $$C = \frac{1020.5624}{100} = 10.2056$$

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2nd-Order Price Change Using Duration + Convexity

- For the 3Y 5% bond at par (P_0=100, y_0=5%):
  - D_{mod} = 2.7232
  - C = 10.2056
- Approximation: $$\frac{\Delta P}{P} \approx -D_{mod}\Delta y + \frac{1}{2}C(\Delta y)^2$$
- +100 bp shock (Δy=+0.01)
  - Duration: −2.7232(0.01)=−0.027232
  - Convexity: 0.5(10.2056)$(0.01)^2$=0.000510
  - Total: $\Delta P/P$≈−0.026722
  - P_1≈97.33

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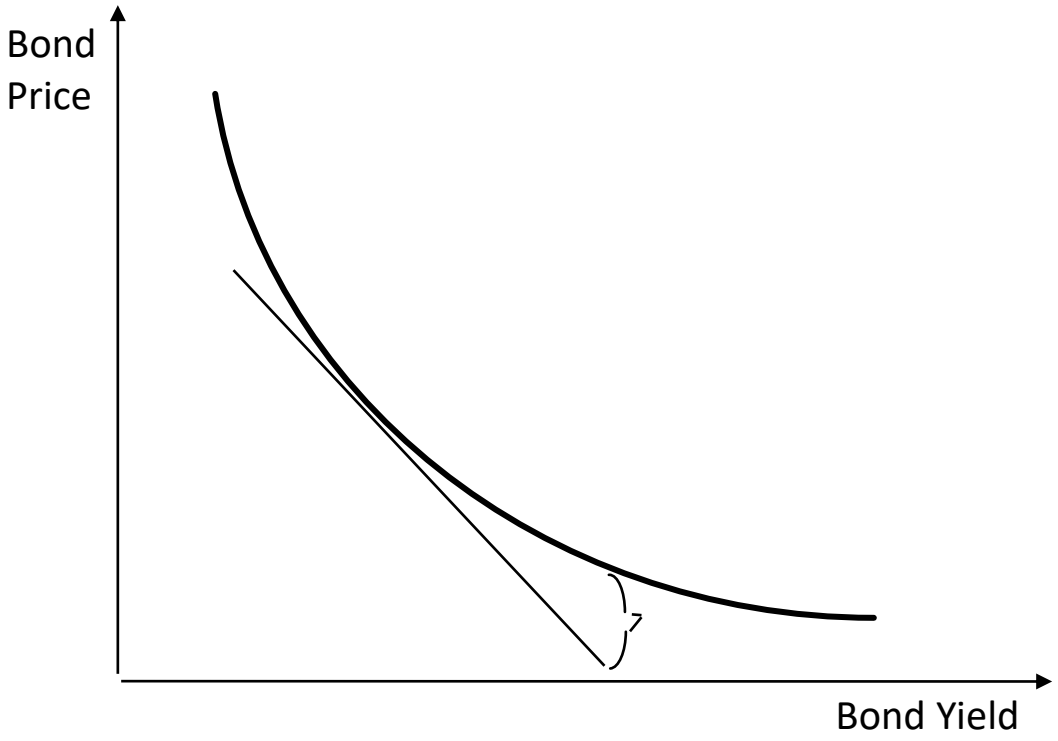
Some Convexity Takeaways

- For equal maturity bonds, zeros have the most convexity.
- Doubling duration more than doubles convexity.
- For a given bond, as yield increases, convexity decreases
- The greater the change in yield, the greater the convexity correction



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Bond Portfolio Risk Management

Example

Bond Portfolio

Par Amt ($MM)	Bond	Price	Dollar Duration
3.0	UST 5yr	99.35	4.144
6.0	UST 10yr	99.19	7.359
-4.0	UST 30yr	92.89	12.896

- Questions:
  - What is the market value of my portfolio?
  - What is my portfolio's interest rate risk and how do I measure it?

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What is my Portfolio's Interest Rate Risk?

Par Amt ($MM)	Bond	Price	Dollar Duration	Market Value ($MM)	DD RISK
3	UST 5yr	99.35	4.144	2.98	124,320
6	UST 10yr	99.19	7.359	5.95	441,540
-4	UST 30yr	92.89	12.896	-3.72	-515,840
				5.22	**50,020**

Weighted Average Duration

- Portfolio Interest Rate Risk
  - Measured using Weighted Average Duration.
  - Weighted Average Duration shows change in Market Value of the portfolio for approximately 1% change in interest rates
  - What assumption does this make about the changing yield curve?

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